

Text Extraction from Image using Deep Learning Technique.



**Under the Guidance of
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Outlines

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- **Literature Survey**
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Problem Statement

- The goal is to develop a deep learning based Optical Character Recognition (OCR) system using a Convolutional Neural Network (CNN).
- The system aims to accurately classify handwritten characters from an image dataset.
- Recognizing handwritten text is more challenging than printed text due to the significant variations in handwritten styles and distortions such as noise.
- The model seeks to improve the accuracy and reliability of handwritten character recognition by using deep learning techniques CNN.

Introduction

- Scene text extraction involves detecting and extracting text from images like street signs, posters, etc.
- It is difficult due to complex backgrounds, lighting variations,, and noise in the images.
- There are some model designed to accurately find and highlight text in images and to overcome the challenges.
- Focuses on segmenting text from images using various methods of Deep learning method of Convolutional Neural Network (CNN) to improve the accuracy of text.

Literature Review

Title	Author	Year	Advantages	Limitations
Accurate, Data-Efficient, Unconstrained Text Recognition with Convolutional Neural Networks	Mohamed Yousef, Khaled F. Hussain, Usama S. Mohammed	2020	It worked very well on both short and long lines of text and also can handle different handwriting styles and sizes.	Existing line recognition methods can't handle paragraphs or multiple lines without line segmentation algorithms.
EPAN: Effective parts attention network for scene text recognition	Yunlong Huang, Zenghui Sun, Lianwen Jin, Canjie Luo	2020	EPAN needs less detailed instructions during its training which is efficient.	The two-stage attention mechanism in EPAN takes more time to train.

Literature Review

Title	Author	Year	Advantages	Limitations
Intelligent character recognition using fully convolutional neural networks	Raymond Ptucha , Felipe Petroski Such , Suhas Pillai, Frank Brockler, Vatsala Singh, Paul Hutkowsi	2019	It predicts both arbitrary symbols as well as words from a lexicon. And Avoids dependency on recurrent neural networks and CTC tuning.	The unconventional use of even convolution filters may limit adoption, and handling larger lexicons requires further investigation.
End-to-End Historical Handwritten Ethiopic Text Recognition Using Deep Learning	Ruchika Malhotra, Maru Tesfaye Addis	2023	The paper is one of the first to focus on recognizing historical handwritten Ethiopic text, helping preserve ancient documents. It uses deep learning to effectively handle complex scripts and enhance text recognition.	The model still struggles with high error rates in recognizing complex handwritten texts. It heavily relies on large datasets, which may not perform well on unseen or diverse text styles.

Literature Review

Title	Author	Year	Advantages	Limitations
Unambiguous Scene Text Segmentation With Referring Expression Comprehension	Zejian Yuan, Chunhua Shen, and Luc Van Gool	2020	The paper combines visual and language features to accurately detect and segment text in busy, cluttered images, improving text recognition.	It process the input text sequentially which slow down the training time as it enables the text to go under process one by one.
Distance Transform based Text-line Extraction from Unconstrained Hand-written Document Images	Suman Kumar Bera, Soumyadeep Kundu, Neeraj Kumar, Ram Sarkar	2021	Placing paragraphs in a one-page document has always yielded impressive results in most cases.	In multi-page documents, if there's a line between paragraphs, the method might mistake it for a new line.

Research Gap

- CNN's automated feature learning significantly performs handcrafted feature extraction in complex text structures.
- Traditional Support Vector Machine(SVM) models face performance limitations, whereas CNNs achieve superior accuracy in real-world scenarios.
- CNNs generalize better for real-world OCR applications, whereas SVMs require extensive preprocessing.
- SVM struggles with real-world text variations, while CNNs offer robust adaptability.
- SVMs are computationally lighter but sacrifice performance, while CNNs optimize recognition accuracy in the long run.
- CNNs provide a more flexible solution for OCR across different languages, scripts, and handwriting.

Proposed Solution

Convolutional Neural Network (CNN):

- CNN is designed to recognize and understand images.
- It takes an image of text, find important features like edges and shapes, understand patterns, and then predict the character or word in the image.
- CNNs have layers that find features, remove unnecessary details, reduce image size, and make predictions.
- After resizing and preprocessing the images, CNN finds the features and remove noise to recognize new characters accurately.

Dataset Description:

The model was trained and tested using Hybrid Character Dataset, which combines the EMNIST (Extended Modified National Institute of Standards and Technology) with a custom-built real world images of our handwritten and printed text images.

No of sample: 857,000 images

Block Diagram

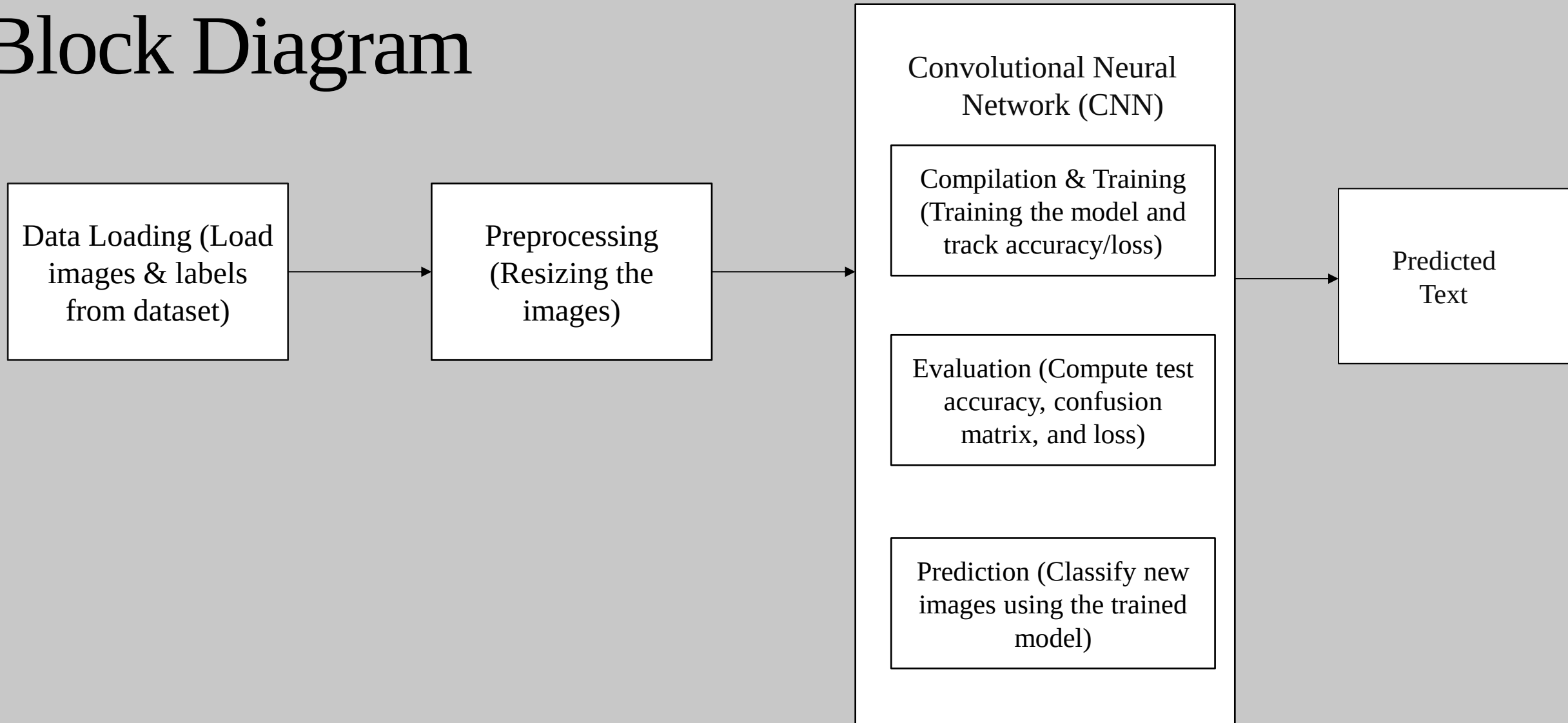


Fig 1: Flow Chart of the Model

Result & Discussion

Table 1: Extracted output of the input images from dataset under CNN

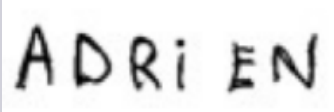
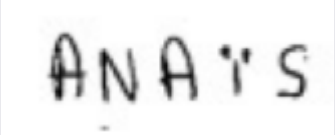
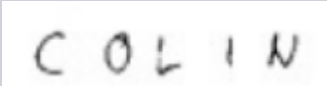
Input Image	Label	Gray Image	Extracted Output
	ADRIEN		
	ANAYS		
	COLIN		
	SARAH		

Table 2: Extracted output of the input images from dataset under CNN

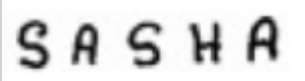
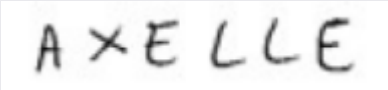
Input Image	Label	Gray Image	Extracted Output
	SASHA		
	EVA		
	AMBROISE		
	AXELLE		

Table 3: Extracted output of handwritten text images from our custom dataset under CNN

Input Image	Label	Gray Image	Extracted Output
	MAGIC		
	BALL		
	CV RAMAN		
	HOLD		

Table 4: Extracted output of printed text images from our custom dataset under CNN

Input Image	Label	Gray Image	Extracted Output
	SAMPLE		
	SMILE		
	EMAIL		
	ANALYSIS		
	PEACE		

accuracy: 0.9236
Precision: 0.9306
Recall: 0.9252
F1 Score: 0.9264

Fig 2: Evaluation of Model Accuracy

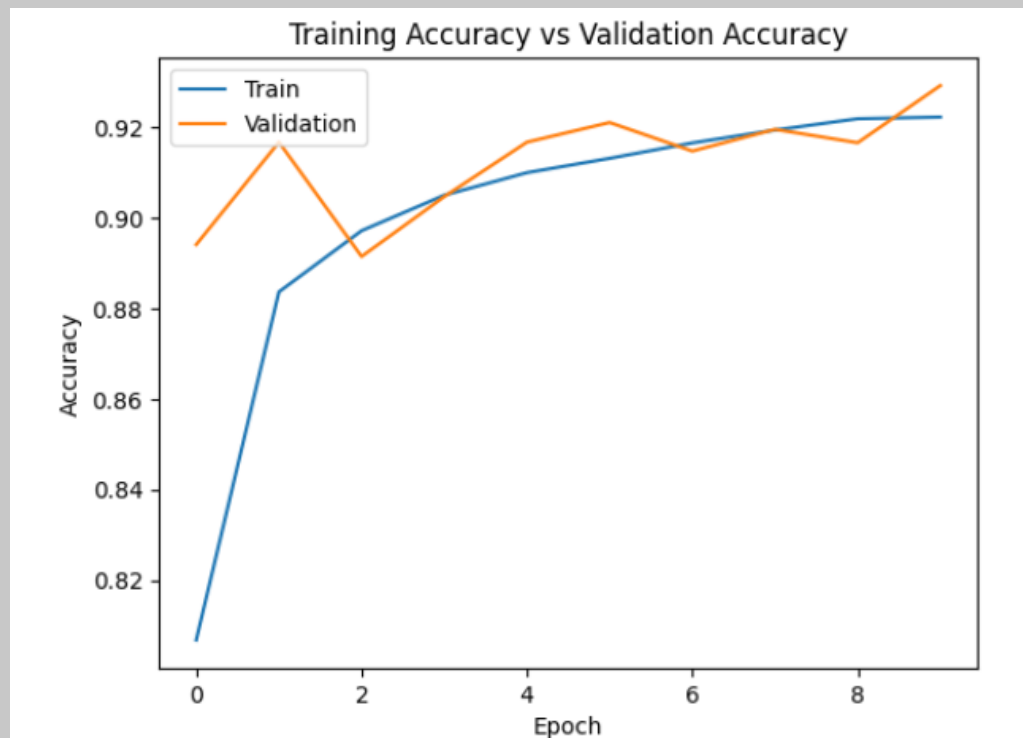


Fig 3: Graph between Training Accuracy and Validation Accuracy

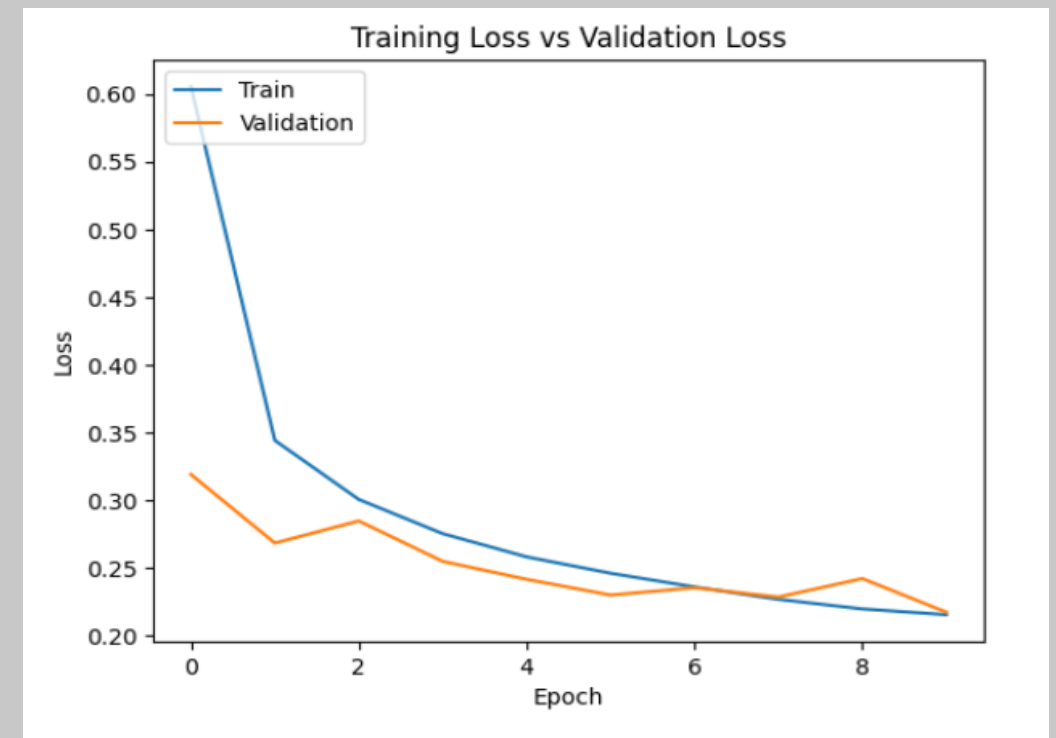


Fig 4: Graph between Training Loss and Validation Loss

Conclusion

- Developed a text extraction solution integrating OCR with CNN-based classification.
- It achieves high accuracy by leveraging convolutional layers for spatial feature learning.
- Reliable performance on any text across varied backgrounds and orientations.
- CNNs have the ability to handle complex text types such as multilingual and stylistic fonts.
- This model can be deployed for real-world OCR tasks, including document scanning and digitization.

References

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